Week 2 | Lab Practice 02

Software Project Management: LaTeX Document Preparation in Overleaf

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Learning objectives

Create a structured LaTeX document using Overleaf; practice titles, sections, subsections, lists and tables; then prepare a one-page SPM project proposal.

Lab rule: Type all commands and content manually. Do not copy and paste code or proposal text.

Step 1. Create an Overleaf project

Visit overleaf.com, sign in, select New Project > Blank Project, name it SPM_Week2_Lab2, and open main.tex. Click Recompile after each exercise.

Step 2. Basic document structure

```
\documentclass[12pt,a4paper]{article}
\begin{document}
Welcome to Software Project Management.
\end{document}
```

Expected output: Welcome to Software Project Management.

Step 3. Document title

```
\title{Software Project Management Proposal}
\author{Team Alpha \\ Zhengzhou University}
\date{\today}
% Insert the next command after \begin{document}:
\maketitle
```

Expected output: A centered proposal title, team/university name and current date. Put title metadata before \begin{document}.

Step 4. Sections and subsections

```
\section{Project Overview}
This project helps students manage assignments.
\subsection{Problem Statement}
Students need a way to track deadlines.
\subsection{Proposed Solution}
The system organizes tasks and reminders.
```

Expected output: 1 Project Overview; 1.1 Problem Statement; 1.2 Proposed Solution, each followed by its paragraph.

Step 5. Bulleted and numbered lists

```
\begin{itemize}
  \item Track deadlines
  \item Assign tasks
\end{itemize}
```

```
\begin{enumerate}
  \item Gather requirements
  \item Develop the prototype
  \item Test the system
\end{enumerate}
```

Expected output: Two bullet points followed by a numbered list of three project stages.

Step 6. Team roles table

```
\begin{tabular}{|l|l|l|}
\hline
No. & Member & Role \\ \hline
1 & Student A & Project Manager \\ \hline
2 & Student B & Developer \\ \hline
3 & Student C & Tester \\ \hline
\end{tabular}
```

No.	Member	Role
1	Student A	Project Manager
2	Student B	Developer
3	Student C	Tester

Step 7. Useful commands

Command	Purpose
\section{} / \subsection{}	Headings
\textbf{} / \textit{}	Bold / italic
\item	List entry
& / \\ / \hline	Table column / row / rule
\maketitle	Display title

Final practice: Complete one-page proposal

Using the commands practiced above, each team must create its own one-page A4 SPM proposal. Include: (1) Project Title, (2) What, (3) Why, (4) How, (5) Who, (6) Expected Outcomes, and (7) Team Members and Roles. Include at least one subsection, a bulleted list, a numbered list and a team responsibilities table.

Complete sample LaTeX document (type manually)

```
\documentclass[10pt,a4paper]{article}
\usepackage[margin=0.7in]{geometry}
\usepackage{enumitem}
\setlist{nosep}
\begin{document}
\begin{center}
{\Large\bfseries Software Project Management Proposal}\\
\textbf{Student Task Management System}\\
Zhengzhou University\\
School of Computer and Artificial Intelligence\\
Instructor: Dr. Hussain
\end{center}
\section*{1. What: Project Description}
A web application to organize academic tasks and deadlines.
\section*{2. Why: Problem Statement}
Students need one place to monitor assignments and group work.
\section*{3. How: Project Plan}
\begin{enumerate}
\item Gather requirements
```

```
\item Design and develop the prototype
\item Test and document the system
\end{enumerate}
\subsection*{Management Approach}
The team will review progress every week.
\section*{4. Who: Target Users}
\begin{itemize}
\item University students
\item Project teams and instructors
\end{itemize}
\section*{5. Expected Outcomes}
A working prototype, tested features and documentation.
\section*{6. Team Members and Roles}
\begin{center}
\begin{tabular}{|l|l|l|}
\hline
Member & Role & Responsibility \\ \hline
Student A & Manager & Schedule and coordination \\ \hline
Student B & Analyst & Requirements \\ \hline
Student C & Developer & Implementation \\ \hline
Student D & Tester & Quality checks \\ \hline
\end{tabular}
\end{center}
\end{document}
```

Expected output: A centered heading followed by labeled proposal sections, a numbered plan, bullet list, management subsection and a bordered team table, fitting on one A4 page.

Final Task and Submission

Complete all lab exercises and compile your LaTeX document successfully in Overleaf.

Save both the PDF (.pdf) and LaTeX source (.tex) files in one folder named with your student ID and name.

Submit this folder to your class monitor before the end of this class.

All LaTeX code must be typed manually. Copying and pasting is not allowed.